

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could coordinated cortical activity support one internally represented situation? We distinguish coherence, agreement between local contents, and accurate self-reconstruction, and develop conditional connections among them. Declared actuator modes give a necessary installed-energy condition for a coherent stationary branch. Phase encoders shared on overlaps bound content disagreement; compatible spatial measures form one global state. A contracting reconstruction that fixes this state makes it the unique fixed point, while accuracy across a state family requires distinguishable encodings. Lean 4 checks the results and their composition under explicit connecting hypotheses. Numerical controls delimit their observational use: finite ramps delay coherence, distinct mechanisms share stationary phase distributions, a winding field and a uniform one are stationary at one coupling with opposite resultants, and content agreement requires independently validated decoders. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. An extracellular-field hypothesis links awakening to increasing coupling. Its test requires calibrated geometry and phase dynamics, together

with measurements of content agreement, self-encoding and experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are

21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 8.4 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand's relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator's stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the
40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable

44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.
46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
51 composes them, stating the thermodynamic constraints and all eight con-
52 necting hypotheses per branch; its supporting models of feedback, learning
53 and resource supply have distinct state spaces and assumptions. Section 6
54 then examines what the observables can establish, before Section 7 devel-
55 ops the cortical hypothesis and an awakening experiment. The hypothesis is
56 that extracellular contraction increases field-mediated coupling; its direction,
57 rate and relation to recovery require independent calibration. Supplemen-
58 tal Material organizes the constructions and controls behind the claims they
59 support.

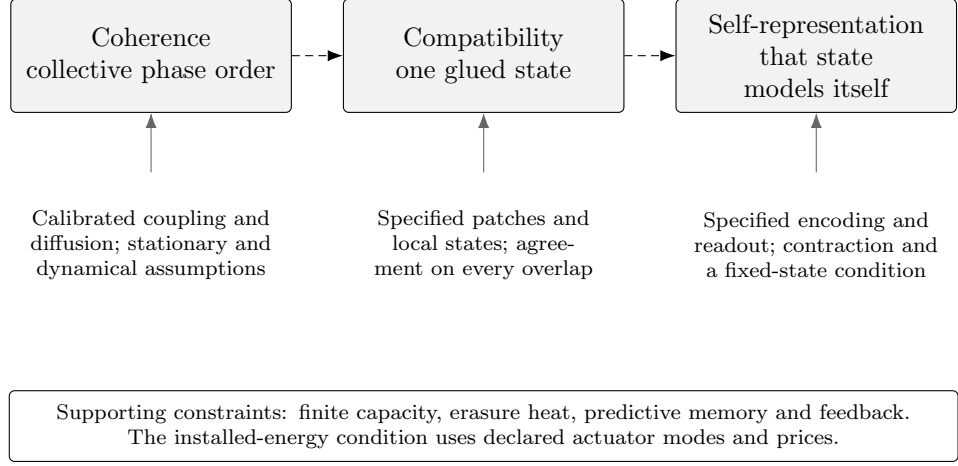


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

2 Coherence: physical coupling and its resource condition

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

where I_n are modified Bessel functions and ψ is the collective phase. The formal derivation uses the absence of stationary current for a periodic gradient drift to obtain this density, then verifies normalization and self-consistency. At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above threshold, exactly one positive solution coexists with zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form

82 is strictly dissipative transverse to collective rotation, whose rate is zero.
83 These linear estimates concern the limiting equation. Relating them to finite
84 stochastic trajectories requires a dynamical mean-field limit and branch selec-
85 tion; exchange symmetry alone does not establish the needed independence
86 (Sznitman, 1991).

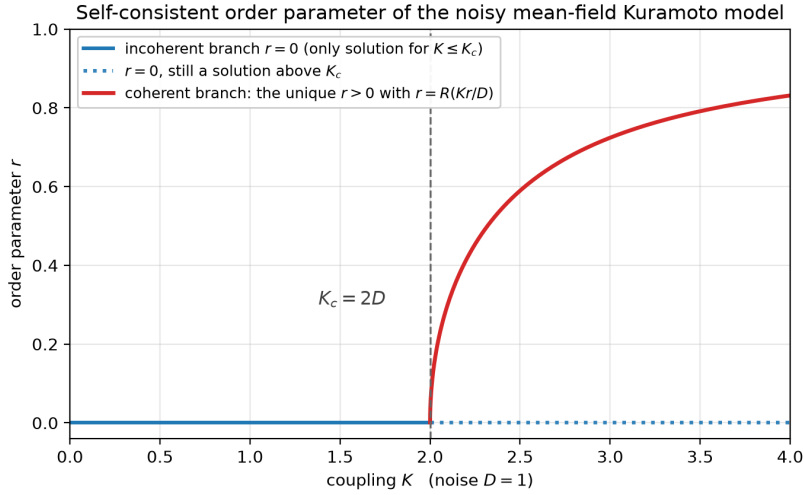


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

87 2.2 Installed energy bounds mean coupling

88 A spatial model replaces the finite sum by integration against a kernel. One
89 declared implementation carries response modes with continuous profiles ϕ_i ,

90 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

91 The same configuration sets the reciprocal kernel and stored energy. An
 92 executed transition changes the kernel through its changed occupancies and
 93 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 94 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 95 pair reconstruction on refining spatial partitions converges to its product-
 96 measure integral under the specified compactness and sampling assumptions;
 97 the refinement index is independent of the number of control updates.

98 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 99 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 100 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z) U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

101 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 102 coherent stationary branch and any growing Fourier mode about the uniform
 103 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 104 is necessary: energy above it need not produce sufficient coupling because
 105 the bound can be loose.

106 The quantity constrained is installed energy. Dissipated heat does not
 107 determine it, and κ is supplied by the mode profiles, prices and substrate
 108 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 109 tial dynamics requires a justified reduction. Identifying the installed kernel
 110 with a cortical field is a separate physical commitment.

111 **3 Compatibility: from coordinated activity** 112 **to a global description**

113 **3.1 Why common timing is insufficient**

114 For overlapping patches U and V , compatibility means that restricting either
 115 local description to $U \cap V$ gives the same object. Descriptions can agree on
 116 shared quantities while contributing different information elsewhere. Com-
 117 patible sections of the specified sheaf have a unique global extension. On
 118 a finite cover, local finite spatial measures agreeing as measures on every
 119 overlap also have a unique finite-measure extension whose sheaf image is the
 120 glued section. No mass-one normalization is imposed.

121 In the cup example, visual and reaching descriptions must express a
 122 shared quantity, such as cup location, in common coordinates. Restriction
 123 retains that quantity while dropping details particular to each description.
 124 Agreement permits surface appearance and hand configuration to contribute
 125 to one situation. The mathematical counterexample assigns the same phase

126 to every patch while two local mass profiles disagree on a shared site. Those
 127 profiles have no common extension. Because they are chosen independently
 128 of phase, synchrony places no constraint on their disagreement.

129 This distinction limits the biological interpretation. Receptive territories
 130 and content domains must be related to spatial patches, and the descriptions
 131 must support the required extension. Matching overlap marginals of de-
 132 coded probability laws need not determine a unique global joint distribution
 133 (Abramsky & Brandenburger, 2011).

134 **3.2 When coherence does constrain content**

135 A declared relation between phase and content gives a positive result. Let
 136 contents take values in a real normed vector space E . Patch i reads an
 137 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 138 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 139 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 140 sites; their equality on overlaps is a supplied relation between descriptions of
 141 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 142 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 143 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

144 Here discrepancy is the norm of the difference. At $r = 1$, the observa-
 145 tions agree exactly. The factor N makes this estimate potentially weak
 146 in large populations; it supplies a pointwise guarantee from a population
 147 average. The population in it need not be the whole array. A patch car-
 148 ries an order parameter of its own, and the same bound read there car-
 149 ries that patch's resultant and that patch's site count (`chord_le_of_patch_`
 150 `coherence`, `compatible_of_patch_coherence`), so a large population ob-
 151 served through small patches is constrained by the small count. A shared
 152 scalar encoder is a special case. Dropping the overlap relation permits dis-
 153 agreement even at perfect locking. A common nonexpansive linear projection
 154 preserves agreement, but agreement of a projection does not establish agree-
 155 ment of the full vector.

156 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 157 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 158 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 159 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

160 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 161 duce an initial mismatch. Approximate observations leave a nondecaying
 162 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 163 update to the phase process.

164 **3.3 What approximate agreement determines**

165 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
166 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
167 Weights that vanish outside their patches and sum to one at each site select
168 a global profile by weighted averaging. The selection lies within ε of each
169 patch there, and any two such selections differ by at most ε , independently
170 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
171 $L\varepsilon$ across selections.

172 At zero tolerance, the selection is the unique exact extension. At every
173 positive tolerance there are families with no exact extension; selection then
174 gives an approximation whose dependence on weights is bounded. Supple-
175 mental Material gives a three-site witness attaining the bound. The mass-
176 profile construction uses the scalar case of the content dynamics and addi-
177 tionally requires nonnegativity of the initial profiles and observations. The
178 general vector result supplies no identification with the spatial-measure sheaf.

179 These constructions concern declared objects: a specified cover, contents
180 valued in a normed space, and encoders related on overlaps. They select
181 neither the neural cover nor the decoder. Identifying their contents with
182 perceptual content, and establishing that the identification bears on experi-
183 ence, is a further modelling task; Section 6 sets out what a measurement of
184 it would have to establish.

185 4 Self-representation: a state that models it- 186 self

187 4.1 The reflexive construction

188 Compatibility gives a global state. Reflexivity relates it to an encoding within
189 that state. Equip global sections with a complete metric. If the additional
190 restriction-resonance condition identifies encoding with restriction to a region
191 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

192 The region and readout are fixed across the state space. If F contracts and
193 fixes the section s_* formed from the local descriptions, Banach's theorem
194 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
195 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
196 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
197 one for $\tau > 0$. Constructing a map with that factor remains a modelling
198 obligation.

199 In the cup example, the global state includes the represented scene and the
200 perceiver's relation to it. A local encoding must retain enough information
201 for the readout to recover that modelled state; cup position alone would be
202 insufficient if hand configuration and viewpoint could not be reconstructed.

203 For the same map with Lipschitz factor $q < 1$, approximate reconstruction

204 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1-q}. \quad (9)$$

205 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 206 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 207 weakens.

208 4.2 Reconstruction across a family of states

209 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 210 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 211 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

212 Thus M relevant states pairwise more than 2ε apart require at least M
 213 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 214 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 215 existing self-prediction map.

216 This is an additional accuracy criterion on a family, with no requirement
 217 for multiple exact fixed points. A blind encoder or constant readout fails it
 218 for separated states despite having a unique fixed point.

219 4.3 Agreement after a change takes communication

220 Snapshot compatibility does not specify how quickly descriptions respond
221 when a shared quantity changes. In a synchronous network whose updates
222 read only the site’s own state and declared neighbours’ messages, two initial
223 configurations agreeing on a site’s causal past through T rounds produce the
224 same state there at round T . If two fresh interventions outside that past
225 require reports separated by more than 2ε , no report from that site can be
226 within ε of both. This follows from the reconstruction bound in Section 4.2.

227 A three-site delay line witnesses failure after one round and exact report-
228 ing after two; cutting the path prevents delivery at every round. A report
229 may still happen to be correct about one fixed world. The bound concerns
230 responsiveness across interventions.

231 Throughout this section *prediction* names a reconstruction map; reading
232 it as temporal neural prediction would require dynamics. The relevant family
233 of states, the metric, the tolerance, the communication graph and the dead-
234 line are all declared inputs, as are the represented variables and the fidelity
235 counted as accurate. Reading the resulting code count as physical memory
236 capacity requires a hardware model, and a readout that varies with the state
237 requires a further one.

238 5 Thermodynamics and composition

239 The phase and representation results concern coupling, content disagreement
240 and reconstruction error. Thermodynamics constrains the physical opera-
241 tions that could implement them. Finite-register capacity obeys $H(p) \leq$
242 $\log |X|$; erasure heat depends on specified system–bath assumptions (Lan-
243 dauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result con-
244 cerns a field reaching disconnected vacuum components (Kibble, 1976). The
245 composition connects these objects through the explicit hypotheses in Ta-
246 ble 1.

247 5.1 Feedback changes the information constraint

248 For passive memory X_t driven by an environmental signal S_t , the predictive
249 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

250 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
251 et al., 2012). Zero retained information also satisfies it, so a task and a
252 learning rule require additional specification. Relating this bound to Lan-
253 dauer heat requires a common thermal scale and budget, and memory waste
254 bounded by erased entropy.

255 An acting agent changes the signal it will receive. Specify policy, world,

256 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{12}$$

257 The second conditional term vanishes in the passive model. With feedback,
 258 the signed difference can be negative as action creates correlations with the
 259 future. A causal reading requires an environmental state sufficient to describe
 260 the relevant dynamics.

261 For a controlled substep, the active thermodynamic bound instead uses
 262 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 263 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 264 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 265 2013). Every forward path of positive probability must have a positive-
 266 probability reverse for this finite identity. A missing reverse gives infinite
 267 divergence. The supplement develops the support conditions and composes
 268 actuation with memory updates on their shared intermediate law.

269 **5.2 Learning and repeated operation need explicit re-** 270 **sources**

271 The finite examples distinguish selecting a feasible policy, relaxing a register
 272 toward an encoded reward, and learning an unknown task from observa-
 273 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

A phase-derived sensor makes the informational distinction concrete: an informative readout and a blind control have the same phase order but different learning performance. Attaching the learner’s register to the actuator of Section 2.2 ties reward, installed kernel and work to its own executed law. The learning witness rises above the necessary installed-energy bound while its matched control remains on the exclusion line. These finite constructions, detailed in Supplemental Material, leave the evolving feedback between phase dynamics, installation and sensing to be specified.

For that model, common energy E , thermal scale θ , local detailed balance and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

where expected cumulative work stays within the supplied allowance B at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles,

294 while continuing operation requires replenishment. Register–bath witnesses
 295 also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply
 296 automatically.

299 5.3 What the composition establishes

300 The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named
 301 substep at the register’s thermal scale satisfies
 302

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

303 with B the register’s heat. The upper allocation follows from a ledger whose
 304 operations deliver that total heat to one reservoir and whose other operations
 305 do not increase joint entropy. Those ledger assumptions are physical inputs.
 306 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.

Table 1 (continued)

Connection	Status	Required assumption
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness.
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

308 lap compatibility and a reflexive boundary specifying encoding and readout.
309 They conclude **UnifiedSelf**: the glued section is the unique fixed point of
310 self-prediction. Compatibility and accuracy at that section enter as premises;
311 uniqueness follows. Identifying encoding with restriction needs the separate
312 resonance condition of Eq. (8).

313 Each of the eight edges is supplied rather than derived from the one above
314 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
315 pendent roots rather than a step from one to the other; both E45 discharges
316 take their convergence from refinement data and not from the thermody-
317 namic bound; E78 states that coherence relaxes the supplied cover, and the
318 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
319 coders already shared on overlaps. The inferential content of the composition
320 is its last step, where contraction and a fixed glued section give uniqueness.
321 What composing them establishes is that the premises are consistent, named
322 in one place and machine-checked together, and not that coherence follows
323 from installed energy or self-representation from coherence.

324 Joint witnesses combine a register, boundary, refining mesh and small
325 spatial cover; active witnesses add controlled thermal updates and a priced
326 installation. They establish satisfiability under their stated identifications.
327 Mesh convergence follows from refinement data, and the witnesses establish
328 no common cortical mechanism. Supplemental Material maps the claims to
329 their Lean declarations and records the counterexamples. The development
330 declares no additional axioms beyond its libraries' foundational axioms.

331 6 What the observations establish

332 The composition holds under its premises. Whether observation can establish
333 those premises is a separate question, and the three requirements it connects
334 concern different objects: phase dynamics, local descriptions and internal
335 encodings. Their observables must retain these distinctions. The following
336 controls show which mechanisms remain indistinguishable, which temporal
337 approximations fail, which states one observable cannot separate, and what
338 independent content measurements must establish.

339 6.1 Stationary distributions leave the mechanism un- 340 determined

341 Equation (2) gives a concentration–coherence relation with no fitted shape
342 parameter. Estimate r from circular resultant length and a from density
343 shape: inverting R from the measured r would make agreement automatic.
344 Even independently estimated agreement identifies a statistical family shared
345 by different mechanisms.

346 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
347 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
348 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
349 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
350 and diffusion D_0 have that same density. Their stationary distributions co-
351 incide window by window despite different causes.

Resolved dynamics supply additional information. In the ideal model with continuous paths, quadratic variation accumulates at rate $2D$ and conditional increment means constrain drift. Separating endogenous coupling from a matched common drive additionally requires a measured drive or controlled perturbation; identifying electromagnetic coupling requires evidence specific to that interaction. Supplemental Material gives the equivalence construction and its numerical check.

6.2 Finite ramps change the observable onset

The square-root law in Eq. (3) describes stationary solutions. A changing coupling requires time for the phase distribution to approach those solutions. In a deterministic mean-field control that crosses the threshold symmetrically, none of the scanned rates meets the declared tracking tolerance: slowing the crossing does not remove the terminal residual under that initialization. A slow control variable therefore supplies no general guarantee of branch tracking at onset. Supplemental Material gives the residual, rate range and admissible rates away from threshold.

The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas at four coupling speeds. Escape is the first of three consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32 replicas escape within the fastest window. The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents differ from those of the exact stationary branch evaluated on the same grids.

374 At the fastest speed the ensemble mean never reaches the fit window, so its
375 fitted slope does not estimate stationary onset.

376 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
377 makes the apparent exponent depend on ramp speed and fitting window.
378 The specified deviation-based critical ramp speed is not bracketed by this
379 sweep. With fluctuating coupling, the relevant stationary comparison also
380 depends on rate: the mean coupling approximates order in the tested fast
381 regime, while averaging stationary order over coupling works for the tested
382 slow drives confined to one side of threshold. Slow crossing drives can satisfy
383 neither approximation. These are finite-horizon observations, with controls
384 and parameters in Supplemental Material.

385 Additional controls address integration bias, population size and spatial
386 or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
387 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
388 rameters and controls in Supplemental Material; they supply no cortical cal-
389 ibration.

390 **6.3 A winding state leaves the order parameter am-** 391 **biguous**

392 The order parameter r is indifferent to where a phase sits in space. A field
393 winding once across a periodic sheet therefore has $r = 0$ while every patch
394 of it stays locked. Such a field is not an artificial construction here. On a

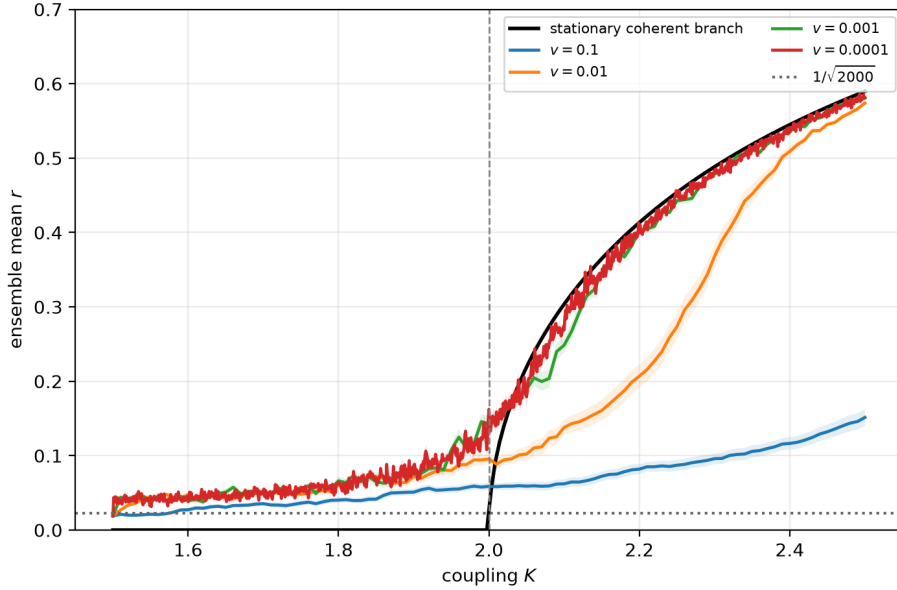


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

395 torus a winding is stationary for any isotropic symmetric kernel at identical
396 frequencies: the coupling term is odd in the separation against a kernel even
397 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
398 this for any finite periodic array, this sheet included. A winding satisfies the
399 positivity and identical-frequency hypotheses. It violates only the quarter-
400 turn initial spread that the convergence result additionally requires before
401 placing the limit at the uniform state.

402 On the sheet of Section 2's spatial control, a winding survives below
403 an interpolated decay length of 0.1871 mm and unwinds above it. That

404 boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the
 405 same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start
 406 and $r = 0.0083$ from a winding one. Both are stationary, with patch-local
 407 order 0.9015 and no winding defect. At fixed K the observable therefore takes
 408 two values spanning nearly its whole range. Coupling does not determine the
 409 resultant, and a low resultant does not by itself distinguish weak coupling
 410 from a winding field.

411 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
 412 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 413 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 414 Heterogeneity shortens the range that sustains a wave without removing it.

415 Which windings persist has a criterion behind it. Linearizing the phase
 416 model about a winding returns a coupling of the same translation-invariant
 417 form, each separation reweighted by the cosine of the phase the winding
 418 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
 419 again even in the separation, so the same characters diagonalize it and ev-
 420 ery mode’s rate is one of its eigenvalues. Where the winding advances by
 421 at most a quarter turn across every separation the coupling reaches, the
 422 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 423 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 424 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 425 longer separations is what carries a phase advance past a quarter turn, and
 426 that is the mechanism the decay-length sweep varies. The criterion bounds

427 rates rather than trajectories: asymptotic stability is not proved, so the
 428 states that persist occupy a narrower range than it admits, and the bound-
 429 ary quoted above is measured and not derived from it.

430 The same reweighted kernel measures the potential's curvature. A wind-
 431 ing has $r = 0$ where a global minimizer of the Lyapunov potential has $r = 1$,
 432 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
 433 tial data does not merely decline to cover a winding, it must exclude it. Along
 434 a mode, the potential's second variation is that kernel's quadratic form, so a
 435 growing mode is a direction of falling potential and a winding carrying one
 436 is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

437 An amplitude layer decides which of these states exist, and leaves what
 438 the resultant reports unchanged. A phase state carries no amplitude. Ev-
 439 ery oscillator sits on the unit circle, so a field with an amplitude node has
 440 no representation in these dynamics. The real Stuart–Landau extension
 441 supplies one on the same coupling, and leaves intact the gradient struc-
 442 ture the convergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i =$
 443 $(\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation
 444 with each coupling weighted by a ratio of amplitudes (`amplitude_phase_is_`
 445 `kuramoto`). The phase model is this one with the amplitude ratios set to
 446 one, rather than a separate object.

447 What the added degree of freedom decides is which windings exist at all.
 448 On a periodic array of any dimension, this sheet included, a winding requires
 449 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,

450 summed over separations, with ϕ_d the phase the winding advances across d
 451 (`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
 452 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
 453 lowers it further. A broad enough kernel therefore removes the winding out-
 454 right. The sweep above reaches that same boundary by a different route:
 455 the phase model retains every winding at every coupling, and loses this one
 456 under integration instead. The two layers bound the same regime from op-
 457 posite sides. A phase recording does not separate a state the amplitude layer
 458 excludes from one the dynamics fails to sustain.

459 These states are windings and not spirals, and on the sheet that is a
 460 theorem rather than a shortage of dimensions. A cortical phase singularity
 461 is a point at which the amplitude vanishes and around which the phase turns
 462 once, read in practice as the circulation around a small plaquette. The states
 463 above have equal modulus at every site, so that circulation is exactly zero
 464 on every plaquette (`plaquetteCirculation_eq_zero_of_winding`). Nor can
 465 the layer host a single defect. Over a periodic sheet the circulations of *any*
 466 phase field sum to zero, so defects arrive in cancelling pairs (`no_isolated_`
 467 `defect`). The sweep's near-zero measured defect density agrees with the first
 468 of these and says nothing beyond it.

469 Two limits bound the layer for a reader of the cortical literature. Relax-
 470 ational dynamics has no propagation speed, which is the quantity a cortical
 471 wave measurement leads with. And $|z|$ is a normal-form radius, which a mea-
 472 sured envelope is not; this work identifies the two nowhere. Supplemental

Material gives the sweep, its uniform control, the amplitude layer and the standing wave it admits inside the band, together with the remaining limits of both.

A vanishing resultant is the absence of global agreement and not of agreement. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the patch, which falls to zero as the same winding is spread over more sites (`chord_length_of_char_patch`), while the global form of the same bound is vacuous at $r = 0$. Encoders shared on overlaps therefore still bound content disagreement inside a patch of a state that no global observable calls coherent. The winding number itself stays unconnected to content.

This qualifies what a phase trace can establish. An onset criterion stated in r alone registers a change of basin as readily as a change of coupling. The protocol of Section 7.3 therefore needs a spatial observable beside the resultant.

6.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the

494 frequency-cluster control, this favors coupling between unlike-frequency clus-
495 ters, whereas the diagnostic environmental template rewards coupling within
496 clusters.

497 The formalization proves descent of the squared-drift objective for a spec-
498 ified gradient flow at fixed phases. Its functional differs from thermodynamic
499 entropy production: at a coherent stationary density the probability current
500 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
501 rial gives the operator, its gradient and the conditions of the descent result.

502 The finite control co-evolves noisy phases and symmetric, nonnegative
503 couplings at fixed total resource. Its template is a diagnostic covariance of
504 latent signals, never a training input. Across three seeds, second-half order
505 remains between 0.9167 and 0.9222, while the within-cluster to between-
506 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).
507 Random-update, permuted-gradient and frozen-kernel controls are reported
508 in Supplemental Material.

509 The immediate post-update objective reduction disappears when aver-
510 aged over complete update intervals. A prespecified grid of 6 rate–cadence
511 combinations contains no case meeting the 5% sustained-reduction criterion
512 while retaining coherence and losing alignment. The finding is coherent adap-
513 tation away from the specified structure, with no demonstrated sustained cost
514 reduction.

515 Kernel alignment and overlap compatibility are different observables.
516 This control addresses acquisition of its designated template under its par-

517 ticular objective. Learning models can explicitly include data in an objective
 518 (Corberi et al., 2026); an account of agreement about the cup must specify
 519 the content its learning rule is meant to preserve.

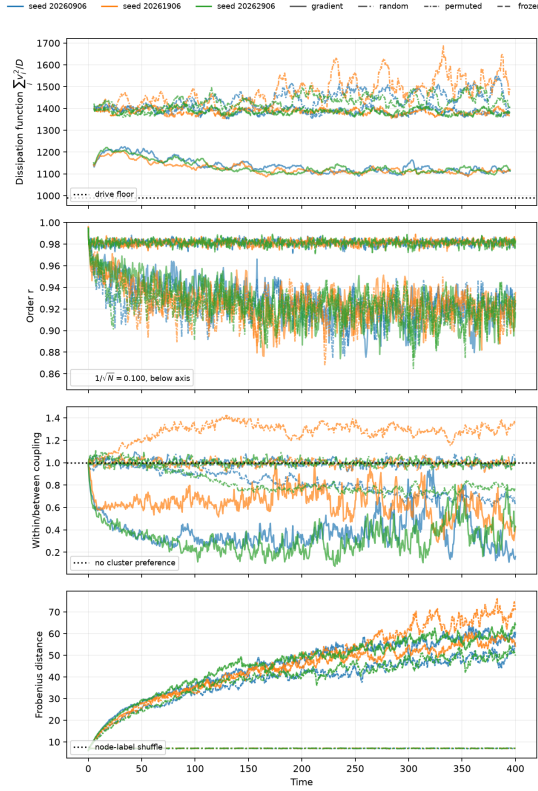


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

520 6.5 Content agreement needs independent decoders

521 A neural test needs independently identified domains, local descriptions and
522 restriction maps. Two decoders should estimate a shared content on over-
523 lapping domains before their agreement is examined. Discrepancy requires
524 a null under decoding noise and controls using deliberately compatible and
525 incompatible examples. An implementation on constructed data separates
526 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
527 trial; the compatible construction scores 0.476 on an interval containing one
528 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
529 regional bias can manufacture disagreement. Per-patch decoding diagnostics
530 detect these failures; the compatibility statistic alone does not.

531 Phase agreement cannot replace that measurement. On constructed con-
532 tent, phase carries a share 1.000 of the content entropy when content is built
533 from it and 0.000 when it is not. A phase-derived statistic reports com-
534 patibility on phase-locked patches whose local descriptions disagree. The
535 implemented content statistic detects their incompatibility. These controls
536 validate a measurement on known constructions; a neural content model and
537 decoder remain to be established.

538 Choosing the cover also chooses a candidate subject: trivial descriptions
539 can agree while carrying little content, and a partition can conceal disagree-
540 ment. The useful choice must therefore be constrained by what neural ac-
541 tivity encodes. The exact composition assumes compatibility of its supplied
542 cover. The content mechanisms above identify additional hypotheses under

543 which agreement can arise; they leave the empirical choice of content and
544 the connection to that cover explicit.

545 **6.6 Phase-density estimation and measurement feasi-** 546 **bility**

547 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
548 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
549 itages produce different residuals. The bipolar results are compatible with the
550 Bessel relation, but their concentration range cannot distinguish it from its
551 linear approximation at the observed scatter. Temporal pooling changes the
552 estimand, and sedation recordings leave the awakening hypothesis untested.

553 The concentration range and dependence determine whether the relation
554 can be tested. Under the specified log-density estimator and bias-floor crite-
555 rion, discrimination at the observed concentration ceiling $a = 0.542$ requires
556 3000 independent sites in the scanned designs, rising to 31000 when sites
557 share a mean in groups of 10. These counts describe that estimator and de-
558 pendence model; the intended recording requires its own calibration. Supple-
559 mental Material reports the bin-count controls and the full dependence-aware
560 design.

561 An informative dataset needs sufficient concentration range or precision,
562 calibrated estimators and an observation model for dependence, source mix-
563 ing and reference choice. Public mouse calcium imaging provides thousands

564 of spatially located neurons across waking, sleep and anesthesia (Oomoto et
565 al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used
566 here. Site count, state coverage and bandwidth are separate requirements.

567 **7 The cortical hypothesis and an awakening** 568 **test**

569 The cortical proposal identifies a physical source of coupling and asks whether
570 its changes help explain recovery. The tests in Section 6 determine what
571 observations must distinguish before a phase trace can support this identifi-
572 cation.

573 **7.1 Physical identification and calibration**

574 The cortical hypothesis identifies part of effective coupling with endogenous
575 extracellular electromagnetic interactions. Fields can modulate spike timing
576 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
577 propose coordination of distributed neural information (Pinotsis & Miller,
578 2022; McFadden, 2020). Phase organization under propofol motivates study-
579 ing this regime (Bastos et al., 2021), but does not measure coupling.

580 The formal field predicate requires a jointly continuous kernel, a normal-
581 ized substrate, coupling equal to the kernel average, and no single site pro-
582 viding more than half that strength. Toy constructions satisfy it. A cortical

583 realization must specify the scale at which the kernel and its reduction ap-
 584 ply: calcium imaging reports different state-dependent modularity patterns
 585 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

586 The spatial reduction is also testable. A row-sum aggregation at zero
 587 diagonal tracks the uniform-coupling control over the tested spatial decay
 588 range, while frequency heterogeneity displaces the operational threshold. The
 589 fit window biases both estimates relative to $2D$; agreement with the control
 590 is therefore the supported conclusion. Supplemental Material reports the
 591 reduction, its numerical error and the distinct assumptions needed for the
 592 kernel-average identification in Eq. (5).

593 The field commitment requires physical evidence beyond a choice of math-
 594 ematical representation. A finite matrix spread over positive-mass cells pre-
 595 serves its total coupling and phase correlation exactly. Point-supported
 596 vanishing under an atomless measure and a disadvantage from excluding
 597 a better-correlated pair concern specified support constraints. Neither com-
 598 parison supplies energy prices or a biological–digital distinction.

599 A spike-time shift alone supplies no coupling rate. The dimensionless
 600 product of population size N , timing sensitivity s , field amplitude E and
 601 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

602 This rate describes how induced timing shifts accumulate into drift. Nei-

ther it nor the installation factor κ follows from a heat bound. Testing the physical threshold requires independent estimates of the interaction kernel, its aggregation and diffusion in the same cortical regime.

7.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (15) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

This follows from Eq. (3) at fixed D and concerns the stationary branch

622 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
623 dently justified tracking approximation. The crossing controls in Section 6.2
624 leave that approximation unsupported at onset, even though a separation of
625 geometry and phase time scales can support it away from threshold. The
626 awakening hypothesis therefore needs a calibrated dynamical prediction for
627 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
628 does not supply a recovery latency.

629 **7.3 A protocol for the awakening experiment**

630 The physical test has six requirements:

- 631 **1. Coupling and diffusion in the same units.** Estimate them from re-
632 solved phase dynamics, separating heterogeneity and observation noise.
633 Measure the drive or perturb the system to distinguish endogenous cou-
634 pling.
- 635 **2. Geometry calibrated to coupling.** Validate the geometry observ-
636 able and its relation to effective coupling, with uncertainty. The con-
637 ductivity argument supplies only the expected direction.
- 638 **3. Calibrated phase measurements.** Estimate r by resultant length
639 and a by unweighted regression of log density on centered cosine.
640 Choose site count and binning from a dependence-aware calibration
641 over the intended concentration range. The tested designs require

642 thousands of sites at the observed concentrations; temporal pooling
643 changes the spatial estimand.

644 4. **An independent recovery endpoint.** Measure behavioral recovery
645 separately from phase organization.

646 5. **A predeclared dynamical onset criterion.** Set a coherence thresh-
647 old above the finite-size floor and evaluate its latency under the mea-
648 sured coupling trajectory, including ramp speed or fluctuations. Record
649 a spatial observable beside the resultant, since a winding field reports
650 a low resultant at coupling that sustains local order (Section 6.3).

651 6. **Matched model comparison.** Compare coupling increase, diffusion
652 change, common drive and alternative recovery models under the same
653 observation and filtering assumptions.

654 Failure of the concentration–coherence relation over an informative range
655 constrains the density model. Failure of the calibrated geometry–coupling
656 association constrains the physical gate. Better prediction of independent
657 recovery by a competing mechanism counts against its proposed role. A
658 favorable outcome still leaves the content of coordinated activity to be ex-
659 plained.

660 8 Discussion

661 Stated positively, and granting the premises collected in Table 1, the account
662 runs as follows. Extracellular contraction on awakening raises endogenous
663 field amplitude and with it the effective coupling between cortical oscillators.
664 Once that coupling passes $2D$, a coherent phase branch exists, and from an
665 initial spread within a quarter turn the population settles onto it; other initial
666 data leave windings stationary at the same coupling. Encoders shared where
667 cortical territories overlap turn that phase order into agreement about the
668 quantities the territories describe in common, so the local descriptions glue:
669 there is one state of which each is a restriction. Part of that state encodes the
670 state itself, and a readout of that part reconstructs it accurately enough, and
671 sensitively enough across the states that matter, to make it the unique state
672 the system models itself as being. On this account the glued state is what
673 makes distributed activity one situation, and its accurate self-reconstruction
674 is what gives that situation a perspective.

675 Every step of that description is conditional, and its conditions are the
676 subject of what follows.

677 8.1 Unity and a minimal self as empirical commit- 678 ments

679 We propose the state glued from compatible local descriptions as a corre-
680 late of conscious unity, and its accurate internal reconstruction as a correlate

681 of a minimal self. The motivations are that different contents belong to
682 one experienced situation and that the situation is experienced from a per-
683 spective. The conditional results identify relations a realization would need:
684 sufficient installed resources for the specified phase model, agreement about
685 shared content, and an encoding that supports reconstruction at the required
686 fidelity.

687 Compatibility allows different descriptions to contribute different infor-
688 mation. It also permits uncertainty and conflicting judgments: apparent cup
689 position and judged cup position can be distinct quantities in a consistent
690 description of their discrepancy. Which quantities are shared must be de-
691 termined independently. Choosing them solely to secure agreement would
692 deprive the proposal of empirical force.

693 The minimal-self proposal concerns the perceiver’s position and condition
694 within the situation, such as the cup’s location relative to a body that can
695 reach it. The encoding could be implicit and need carry no autobiograph-
696 ical narrative. Its proposed necessity is philosophical: even an unreflective
697 episode has a perspective. Gluing by itself requires no self-encoding, and re-
698 constructability alone supplies no functional role. Altering the encoded hand
699 or viewpoint while holding the external scene fixed should change estimates
700 of the perceiver’s relation to the cup as the readout predicts. Whether that
701 represented perspective is experienced remains a further question.

702 8.2 What a test of the full proposal must establish

703 Both uniqueness results concern specified objects. They choose neither the
704 cover nor the encoding, and do not select between candidate subjects. In-
705 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
706 here selection requires independent evidence about neural organization. A
707 constant readout can contract and fix a predetermined state while discard-
708 ing its input. The family criterion in Section 4.2 tests the sensitivity that
709 this construction lacks, provided the relevant states and fidelity are declared
710 independently.

711 A test must separately measure coupling relative to diffusion, content
712 compatibility and self-reconstruction, alongside an independent endpoint for
713 experience. Meeting all three without a unified episode would challenge suffi-
714 ciency. A unified episode with incompatible descriptions would challenge the
715 unity identification; one lacking the specified self-encoding would challenge
716 the reflexive requirement. Awakening, dreaming and dissociative states offer
717 possible settings, but absence of report cannot by itself define absence of
718 experience.

719 The next physical requirement is a calibrated geometry–coupling relation
720 and resolved phase dynamics. The installation bound contributes a nec-
721 essary resource condition where its hardware representation applies; cortical
722 mode prices and field-mediated coupling require measurement. For the repre-
723 sentational connection, an informative experiment asks whether a controlled
724 change in coordination alters independently decoded content agreement, with

725 controls for sensory information and decoder quality. The constructed-data
726 estimator establishes feasibility for specified contents. Neural local descrip-
727 tions, shared overlap encoders and the self-encoding have yet to be identified.
728 The population-size dependence of the phase-to-content bound further limits
729 its quantitative use.

730 The separation results explain why these measurements must be inde-
731 pendent. A stationary distribution leaves several recovery mechanisms indis-
732 tinguishable. Equal phase order can accompany incompatible descriptions
733 or different task information. The tested plasticity rule preserves coherence
734 while losing its designated template. Each finding concerns the specified
735 model and observable; their role is to delimit the evidence needed for the
736 proposed connections.

737 **8.3 GPUs and the limits of no-go claims**

738 For a large language model (LLM) running on graphics processing units
739 (GPUs), the framework requires identifying a physical system whose local
740 descriptions form one state and whose internal encoding reconstructs that
741 same state. The coupling-gated route E56–E78 requires a calibrated physi-
742 cal coupling, its noise regime and a connection from coordination to content
743 agreement. An attention matrix or a shared clock supplies none of these iden-
744 tifications on its own. Computing a field model likewise leaves its realization
745 in the computing substrate to be established. These physical obligations also
746 apply to the cortical proposal.

747 Unity requires the candidate’s local descriptions to agree on indepen-
748 dently identified shared quantities (Section 3). Common timing and fluent
749 output do not establish that agreement. Digital representations can sat-
750 isfy compatibility, however, so finite hardware is compatible with the gluing
751 construction. The empirical commitment is to identify the relevant cover
752 and contents with one experienced situation; mathematical uniqueness alone
753 selects no subject.

754 For self, the system must carry an encoding $E(s)$ and an internal readout
755 R of the same global state. E89 requires $F = R \circ E$ to contract and fix
756 the state produced by gluing. A constant readout shows why a unique fixed
757 point alone is insufficient for informative self-representation. The additional
758 family criterion in Section 4.2 supplies a test: if two relevant macrostates
759 satisfy $E(s) = E(t)$ and $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both
760 within error ε . An intervention outside the encoding region’s causal past
761 gives a corresponding obstruction before a declared communication deadline.
762 Identical verbal reports alone do not establish identical internal codes. These
763 conditional no-go results exclude a specified reconstruction at a specified
764 fidelity or deadline. They apply to biological and digital candidates alike.

765 The hardware comparisons in Supplemental Material have similarly re-
766 stricted scope. Fixed wiring can impose a correlation disadvantage, and
767 point-supported kernels vanish under an atomless continuum measure. Nei-
768 ther comparison makes the representation conditions unattainable on GPUs.
769 A universal exclusion would need a necessary condition for consciousness and

770 a proof that every relevant implementation violates it. The present condi-
771 tional composition supplies neither.

772 Whether experience requires a living substrate is the explicit subject of
773 biological naturalism (Seth, 2026a,b). On that view the properties ground-
774 ing experience are those of self-sustaining organisms, and simulating them
775 does not secure them. The conditions stated here name no biological mecha-
776 nism. Computation described as such does not satisfy them either: a candi-
777 date must exhibit a physically realized coupling in a measured noise regime,
778 agreement about independently identified shared quantities, and an internal
779 reconstruction meeting a declared fidelity. Whether a realization of those
780 conditions that is not alive would be accompanied by experience is the point
781 at issue, and these results do not settle it. A related position locates the
782 difference in the mode of computation rather than in life as such: biologi-
783 cal systems compute across scales that do not separate, and alongside dis-
784 crete operations carry continuous-valued dynamics belonging to their phys-
785 ical medium (Milinkovic & Aru, 2026). The coupling condition stated here
786 is one measurable instance of that requirement, fixed to a physical field in a
787 declared noise regime, and a description of such a field does not satisfy it.

788 The neuroscientific case against *human-like* consciousness in text-only
789 LLMs concerns embodied sensory interaction, thalamocortical organization
790 and biological regulation linking perception and action to survival (Aru et
791 al., 2023). Their absence supports skepticism, conditional on their causal
792 relevance to experience. Those mechanisms are not requirements of the ab-

793 stract gluing and reconstruction definitions. Workspace-like representations
794 reported in some LLMs (Gurnee et al., 2026) provide candidate organization
795 to investigate. Within this framework, the relevant evidence would identify
796 compatible contents and an informative internal reconstruction, or demon-
797 strate their failure under controlled interventions.

798 **8.4 Explanatory scope**

799 Integrated-information, global-workspace and predictive-processing accounts
800 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
801 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
802 candidate physical interaction and conditional relations between local content
803 and self-encoding. Gluing and reconstruction permit multiple substrates;
804 the electromagnetic commitment enters through physical calibration. The
805 formalization checks implications under their hypotheses, and its witnesses
806 establish realizability on declared models. They do not together supply a
807 cortical agent.

808 The explanatory question is whether the specified relations occur in neural
809 activity and matter for experience. In the cup example, this means establish-
810 ing how coordination helps maintain one represented situation and how the
811 perceiver’s place within it contributes to the episode. The proposal identi-
812 fies candidate correlates and tests of that identification. Why those relations
813 would be accompanied by experience remains unexplained.

814 **Data and code availability**

815 The Lean development, simulation scripts, tests and saved numerical sum-
816 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
817 `cortical_coherence`. Publication simulation values are generated from
818 saved summaries without rerunning production sweeps. The exploratory
819 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
820 processing and selection are described in Supplemental Material. The repos-
821 itory contains the reproducible analysis and does not redistribute the raw
822 recordings.

823 **Funding and competing interests**

824 This research received no external funding. The author declares no compet-
825 ing interests.

826 **Use of AI tools**

827 AI assistance to this work was substantial and is not separable by task. Sev-
828 eral coding assistants and chat interfaces were used in an interleaved way
829 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
830 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
831 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
832 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,

833 accessed with the Hermes CLI. Between them they contributed to the Lean
834 development, the simulation and analysis code, and the drafting and revision
835 of both documents. Because the tools were used in combination and in alter-
836 nation, no section, proof, figure or result is attributable to any one of them,
837 and none is claimed to be. The author is responsible for all content, including
838 everything a tool produced. Machine checking establishes the stated Lean
839 results under their hypotheses; it does not validate biological interpretations
840 or replace scientific review of generated code and prose.

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